

CALCULATOR PROJECT REPORT

Project Title

Command-Line Calculator Using Python

Objective

The objective of this project is to develop a simple command-line calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator accepts user input, processes mathematical calculations, and displays the result accurately.

Introduction

A calculator is one of the most fundamental applications in programming. It helps users perform mathematical operations quickly and accurately. This project demonstrates the use of Python programming concepts such as variables, functions, conditional statements, loops, and exception handling.

Problem Statement

Performing calculations manually can be time-consuming and prone to errors. A simple calculator application can automate arithmetic operations and provide instant results while handling invalid user inputs effectively.

Technologies Used

- Python 3.x
- Built-in Python Functions

Features

- Addition of two numbers
- Subtraction of two numbers
- Multiplication of two numbers
- Division of two numbers
- User-friendly command-line interface
- Input validation
- Error handling for invalid inputs
- Protection against division by zero
- Option to perform multiple calculations without restarting the program

System Requirements

Hardware Requirements

- Computer/Laptop

- Minimum 2 GB RAM

Software Requirements

- Python 3.x
- PyCharm / VS Code / IDLE

Methodology

1. Display a menu of arithmetic operations.
2. Allow the user to select an operation.
3. Accept two numbers as input.
4. Perform the selected calculation.
5. Display the result.
6. Handle invalid entries and division-by-zero errors.
7. Ask the user whether to continue.
8. Repeat or terminate the program.

Algorithm

1. Start the program.
2. Display calculator menu.
3. Select operation:
 4. Addition
 5. Subtraction
 6. Multiplication
 7. Division
8. Enter two numbers.
9. Perform calculation based on user choice.
10. Display the result.
11. Ask the user if they want another calculation.
12. If yes, repeat the process.
13. Otherwise, exit the program.

Working

The calculator receives user input through the terminal. Based on the selected operation, the program performs the corresponding arithmetic calculation. Exception handling ensures that invalid inputs and division-by-zero errors are managed properly, preventing program crashes.

Output

Sample Output

Calculator Menu

1. Addition

2. Subtraction

3. Multiplication

4. Division

Enter your choice: 1

Enter first number: 25

Enter second number: 15

Result: 40

Do you want to perform another calculation? (y/n): n

Thank you for using the Calculator!

Advantages

- Easy to use
- Fast calculations
- Accurate results
- Handles invalid inputs
- Suitable for beginners learning Python

Limitations

- Supports only basic arithmetic operations
- Command-line interface only
- No scientific calculations

Future Enhancements

- Graphical User Interface (GUI)
- Scientific calculator functions
- Calculation history
- Memory operations
- Keyboard shortcuts
- Advanced mathematical operations

Conclusion

The Calculator project successfully performs basic arithmetic operations and demonstrates fundamental Python programming concepts. It is a useful beginner-level project that improves understanding of user input handling, conditional statements, loops, and exception handling.

References

1. Python Official Documentation
2. Python Tutorial Documentation
3. Python Built-in Functions Documentation
4. Basic Arithmetic Operations in Python